

KADIYALA RAJESH

Vijayawada, Andhra Pradesh, India | +91 8688053186
| rajeshkadiyalaaa@gmail.com | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

AI & Machine Learning graduate with hands-on experience building and deploying intelligent systems across Computer Vision, NLP, and Generative AI. Proven ability to take projects end-to-end — from model training to production deployment — including a real-world AI-assisted POS system handling 200–300 daily transactions. Strong foundation in Python, Deep Learning frameworks (PyTorch, TensorFlow), and full-stack AI application development using Flask, FastAPI, and React. Passionate about solving business problems with practical AI, with a focus on Computer Vision and Generative AI.

PROFESSIONAL EXPERIENCE

Operations & E-Commerce Associate / AI Tools Developer

Apr 2025 – Mar 2026

LeFrut – Fruit Retail & E-Commerce · Vijayawada, AP

- Designed and deployed an AI-assisted POS billing system (React + Electron) processing 200–300 daily transactions, reducing checkout time by ~30% and streamlining store operations end-to-end.
- Built inventory management module with real-time stock tracking, CRUD operations, and bulk CSV import for ~50 SKUs; integrated thermal receipt printing (58mm/80mm) and keyboard-driven workflow for faster retail throughput.
- Developed a tracking daily sales, expenses, profit/loss, and top-selling products, enabling data-driven restocking and pricing decisions on an avg. daily revenue of ₹20,000.
- Managed Shopify-based online order processing, online payment settlements, and daily financial reporting; maintained operational data records across store and e-commerce channels.

PROJECTS

Deep Learning-Based Dental Caries Detection System | Python · Mask R-CNN · ResNet-50 · BERT · Flask

- Built a medical image segmentation pipeline using Mask R-CNN (ResNet-50 backbone) on the DC1000 dental X-ray dataset (~1,000 images across Normal, Moderate, and Severe caries classes), achieving 80–88% accuracy and mAP of 0.60–0.75.
- Integrated BERT-based NLP module to generate personalized dental care recommendations alongside cavity detection results, creating a dual-output diagnostic system.
- Applied transfer learning, data augmentation, and class-weighted loss functions to handle class imbalance in the dataset; deployed as a responsive Flask web app with real-time detection visualization and severity classification.

Arbitrary Image Style Transfer using CycleGAN + AdaIN | Python · PyTorch · CycleGAN · AdaIN · Flask

- Implemented a custom CycleGAN architecture with Adaptive Instance Normalization (AdaIN) enabling arbitrary style transfer without retraining — trained on ~7,400 unpaired images (trainA: 1,072; trainB: 6,287) across two visual domains.
- Enhanced style-content disentanglement using dynamic feature alignment, residual blocks, and LPIPS-based perceptual evaluation (PSNR, SSIM, FID framework); significantly improved visual fidelity over baseline CycleGAN.
- Built a Flask web application with adjustable style weight controls and real-time image upload interface for user-defined style transfer.

MCQs Generator using NLP & Transformer Models | Python · T5 · Hugging Face · NLP · Flask

- Engineered an intelligent MCQ generation pipeline using T5 transformer model — accepts PDF/text input, applies tokenization, sentence segmentation, and keyword extraction to auto-generate context-aware questions.
- Implemented distractor generation using semantic similarity and linguistic pattern matching to produce challenging, meaningful incorrect answer choices.
- Deployed as a Flask web application with document upload interface for real-time MCQ output; designed for integration with educational platforms and e-learning systems.

AI Writing Assistant | Python · React · Node.js · Flask · OpenRouter LLMs · NLTK

- Developed a full-stack AI writing assistant integrating Large Language Models via OpenRouter API for text generation, rewriting, grammar correction, and tone transformation with sub-second latency per query.
- Created RESTful APIs (Flask + Express) supporting real-time interaction, document generation, and multi-format export (PDF, DOCX, Markdown); managed versioned document data with SQLite.

TECHNICAL SKILLS

Languages & Query: Python, SQL, C++

AI / ML Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, OpenCV, SpaCy, NLTK, YOLO, Mask R-CNN, EfficientNet

Generative AI: LLMs, Prompt Engineering, CycleGAN, AdaIN, OpenRouter API

Data & Visualization: Pandas, NumPy, SciPy, Matplotlib, Seaborn

Web & API Development: Flask, Streamlit, REST APIs

Databases & Cloud: MySQL, Google BigQuery, Google Cloud Platform, Supabase

Dev Tools & Deployment: Git, GitHub, VS Code, Cursor, Google Colab, Kaggle Notebooks, Vercel, Lovable, Bolt, Google AI Studio, Manus

Core Competencies: End-to-end ML pipelines, Model deployment, Data storytelling, Problem-solving

EDUCATION

B.Tech — Artificial Intelligence & Machine Learning

Usha Rama College of Engineering and Technology, Andhra Pradesh

Dec 2021 – May 2025

CGPA: 7.7 / 10

CERTIFICATIONS

- Google Data Analytics Professional Certificate — Coursera
- Data Visualisation: Empowering Business with Effective Insights (Forage)
- Robotics and Controls Job Simulation — Johnson & Johnson (Forage)

AREAS OF INTEREST

Computer Vision | Generative AI | Large Language Models | Natural Language Processing | AI for Healthcare | AI Agents | Prompt Engineering | AI Research